

Vashasetu (ভাষাসেতু)

A Regional Bangla Dialect Translator and Learning App

Software Test Report

CSE 416 – Software Project Design and Development

Department of CSE | Autumn 2026

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Date: 31.10.2026

CSE 416 – Software Project Design and Development (Lab)

Software Test Report

Test plan, test cases and results (based on ISO/IEC/IEEE 29119-3)

Project	Vashasetu (ভাষাসেতু): A Regional Bangla Dialect Translator and Learning App
Team ID	Section: 7B1
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Submitted	31 October 2026

Document Control

Repository	https://github.com/Muskan37/Vashasetu
Version tested	Build of 31 October 2026 (VashaSetu_FULL_latest.zip, service-worker cache v47)
Where it runs	XAMPP on the laptop (http://localhost/vashasetu); shared to phones through a Cloudflare tunnel
Report version	1.0
Prepared by	Shobnom Sultana Muskan, Sadia Akter Liza

Version	Date	Changes
1.0	31 Oct 2026	First version: system, UI, security, load, reliability and accessibility suites, PHPUnit suite (unit, data, integration, HTTP), coverage report, generated-test review, mutation testing, defects D1–D38, native-speaker reviews of the Noakhali, Garo and Chattogram word lists

1. Introduction

1.1 Purpose

This report covers how we tested Vashasetu at the build of 31 October 2026: what we tested, how, the results, and whether it is ready to be released.

1.2 Product overview

Vashasetu translates between Standard Bangla, English and eleven regional Bangla dialects (and Chakma, Marma and Garo), shows a word box with the meaning and reliability of every word, and teaches a dialect step by step. Input can be typed in Bangla, Banglish or English, spoken, or scanned from a photo. It is an installable web app (PWA) in vanilla JavaScript with a PHP 8 API and a MySQL database, run on XAMPP.

1.3 Scope

In scope	Requirements	Levels
Translation, exact sentences, word boxes	FR1, FR3, FR4	Unit, Integration, System, E2E
Bangla, Banglish, English, voice and photo input	FR2	Unit, Integration, E2E
Words dictionary, edits, edit log, admin undo	FR5, FR6	Integration, System, E2E, Reliability
Learning path and quizzes	FR7	Integration, System, E2E
Accounts, history, profile, word of the day	FR8	Integration, System, E2E
Phone use, install, offline, accessibility	NFR1	E2E, Accessibility
Speed and many users	NFR2	Performance, Reliability
Security	NFR3	Unit, Integration, Security
Honest labelling and quality of the word data	NFR4	Data, Integration

Out of scope	Why
Linguistic quality of every dialect word	Needs native speakers. Reviewed so far: the Noakhali, Garo and Chattogram word lists. We test that reviewed words are used and labelled correctly.
Google text-to-speech download	The test computer has no internet. We test the message the app shows instead.
Safari and Firefox	Only Chromium is installed on the test computer.

1.4 References

Document	Version / Link
SRS and design (CSE 355 / CSE 416 project report)	v1.0, 27 Oct 2026
Lab handout: Software Testing and Quality Assurance (CO3)	Steps 1–4
ISO/IEC/IEEE 29119-3; ISO/IEC 25010; OWASP Top 10; WCAG 2.1 AA	2021 / 2018 editions

1.5 Abbreviations

EP: Equivalence Partitioning.

BVA: Boundary Value Analysis.

AAA: Arrange, Act, Assert.

E2E: End to End (real phone-sized browser).

RTM: Requirement Traceability Matrix.

p95: 95th percentile response time.

PCOV: PHP line-coverage driver.

2. Test Environment

OS	Linux test computer (2 CPU cores); the app normally runs on Windows with XAMPP
Runtimes	Apache 2.4.58 (prefork, 150 workers) + mod_php 8.3 with PCOV 1.0 for coverage; MariaDB 10.11 set up like XAMPP (root without password, display_errors On); Python 3; Node 20
Test database	vashasetu, loaded from the CSV data with setup.php?reset=1 before every full run
Browser	Chromium (Playwright 1.56) as a 390 × 844 px phone, Bangla locale; 320–1366 px for layout checks
Test tools	PHPUnit 9.6.17, php-code-coverage 9.2, Playwright, axe-core 4, requests; our scripts: tests/*.py, tests/mutation_check.py, tests/run_coverage.sh

2.1 Test data

Every full run starts from a fresh database loaded from the committed CSV files. Tests create their own accounts (…@example.com), words and history entries with a random code, and remove them again; database changes made by the in-PHP integration tests run inside a transaction that is rolled back. Expected values come from native-speaker sentences and the native-speaker-reviewed word lists, never from the code's own output.

2.2 How to run

- `Start Apache and MySQL in XAMPP and open http://localhost/vashasetu/setup.php?reset=1 once`
- `composer require --dev phpunit/phpunit:^9.6 (once)`
- `vendor\bin\phpunit (Unit, Data, Integration and HttpFlows suites)`
- `python tests/system_tests.py / ui_tests.py / security_tests.py / load_tests.py / reliability_tests.py / a11y_perf_tests.py`
- `python tests/mutation_check.py (planted-bug check of the assertions)`
- `bash tests/run_coverage.sh (everything with coverage, merged HTML report)`

3. Test Strategy

Level / type	Objective	Technique	Tool
Unit	Password, dialect and address rules, Bangla text helpers, phonetic matching, English grammar, Noakhali rule	EP, BVA, AAA	PHPUnit
Data	Word lists: complete rows, valid Bangla script, no duplicates	Rule check over every row	PHPUnit
Integration	Translation pipeline and dictionary with the real database; every API endpoint over HTTP	EP, BVA, error guessing, use cases	PHPUnit
System / E2E	All features as a user uses them on a phone-sized browser	Use cases	Python, Playwright
Regression	Full suite after every fix; one test per fixed defect	Re-run	phpunit, CI
Performance	Speed with 1–300 users, spike, soak, page speed	Load test	load_tests.py, Chrome metrics
Security	OWASP Top 10 that apply: access control, injection, auth, CSRF, headers	OWASP checklist, fuzzing	security_tests.py
Reliability	Parallel edits, database stop/start, restart	Fault injection	reliability_tests.py
Accessibility	10 screens, keyboard, zoom, screen sizes	WCAG audit	axe-core
Test quality	Do the assertions catch real bugs?	Mutation testing (24 planted bugs)	mutation_check.py

Why these tools: the API is plain PHP, so PHPUnit fits, and calling it over HTTP tests routing, sessions and SQL too. Playwright drives a real Chromium on a phone screen. PCOV records coverage both in PHPUnit and in every request the web server answers, so the Python suites count towards coverage as well.

3.1 Entry criteria

- All PHP, JavaScript and Python files pass the syntax check (code_scan.py CS-01–CS-03)

- Every requirement has acceptance criteria in the SRS
- The environment and test data are ready (section 2)

3.2 Exit criteria

Criterion	Target	Result	Met?
Requirements with a test	100%	100% (12 of 12)	Yes
P1 test cases passing	100%	100% (59/59)	Yes
Line coverage, core logic	60%	90.9% (whole PHP code 90.1%)	Yes
Planted bugs caught (mutation score)	80%	23/24 (96%)	Yes
Open S1 / S2 defects	0	0	Yes

4. Traceability Matrix

Built from the test catalogue in section 5: each test case names the requirements it checks, and the status comes from the actual run. Requirements are those of our SRS.

Req	Requirement	Test case IDs	Status
FR1	Translate Standard ↔ dialect, dialect ↔ dialect, English	TC-U03, TC-U09, TC-U10, TC-U11, TC-U17 +14	Pass
FR2	Bangla, Banglish, English, voice and photo input; auto-detect	TC-U06, TC-U07, TC-U12, TC-U13, TC-U14 +8	Pass
FR3	Exact sentence first, word by word only when allowed	TC-U08, TC-U10, TC-01, TC-05, TC-09 +2	Pass
FR4	Word box with meaning and reliability	TC-06, TC-35, E2E-01, E2E-05	Pass
FR5	Dictionary: search, browse, add, edit, delete	TC-08, TC-09, TC-19, TC-20, TC-21 +4	Pass
FR6	Edit log by address; administrator undo	TC-32, REL-01	Pass
FR7	Learning path and quizzes	TC-25, TC-36, TC-37, TC-34, E2E-03 +1	Pass
FR8	Accounts, history, profile, word of the day	TC-U01, TC-11, TC-12, TC-13, TC-14 +5	Pass
NFR1	Phone use, installable, offline start, accessible	TC-36, E2E-04, A11Y-01, A11Y-02	Pass
NFR2	Fast; works with many users	PT-01, PT-02, PT-03, PT-04, REL-02	Fail
NFR3	Security: hashed passwords, prepared SQL, limits, sessions, protected files	TC-U01, TC-U02, TC-U04, TC-U05, TC-10 +11	Pass
NFR4	Unverified meanings labelled; reviewed data is correct	TC-D01, TC-D02, TC-D03, TC-D04, TC-D05 +3	Pass

Requirements	12 (8 FR, 4 NFR)
With at least one test	12 (100%)
With a negative test	8 (67%)

5. Test Cases Specification

5.1 Test case list

84 test cases. Each one is a PHPUnit test method (often with a data provider of several equivalence classes or boundary values) or a group of cases from the earlier suites; "Runs" is the number of individual checks behind it.

TC ID	Req ID	Title / Test Case	Level	Priority	Runs	Result
TC-U01	FR8, NFR3	Password length boundaries: 0, 7 and 201 characters refused; 8, 9, 199 and 200 accepted	Unit	P1	16	Pass
TC-U02	NFR3	Every refused password gets a readable reason	Unit	P2	1	Pass
TC-U03	FR1	Only the 11 supported dialects are accepted (exact names)	Unit	P1	16	Pass
TC-U04	NFR3	Visitor address from the tunnel header is trusted only from this computer	Unit	P1	5	Pass
TC-U05	NFR3	HTTPS is detected directly, behind a proxy and through the tunnel	Unit	P2	7	Pass
TC-U06	FR2	Bangla text is normalised (spaces, punctuation, joiners, two spellings of য় and ড়)	Unit	P1	16	Pass
TC-U07	FR2	ো typed as ে + া is the same letter as ো	Unit	P1	6	Pass
TC-U08	FR3	Verb spelling variants at the 2/3-letter boundary (খাবো / খাব)	Unit	P2	6	Pass
TC-U09	FR1	English contractions are expanded (can't, won't, curly apostrophe)	Unit	P2	6	Pass
TC-U10	FR1, FR3	Text is split into sentences; Mr./Dr. and list numbers do not end a sentence	Unit	P1	14	Pass
TC-U11	FR1	The input's punctuation is put back into the translation	Unit	P1	23	Pass
TC-U12	FR2	Banglish full stops become দাঁড়ি; ... and 2.5 stay	Unit	P2	5	Pass
TC-U13	FR2	Roman typing of a Bangla word matches it (ami = আমি); different words do not	Unit	P1	5	Pass
TC-U14	FR2	Phonetic edit distance: boundaries, symmetry and costs	Unit	P2	8	Pass
TC-U15	FR2	Roman letter rules (chh, x, f, w, y, doubled letters)	Unit	P2	8	Pass
TC-U16	FR2	Fuzzy look-up keys never contain the empty key (1-letter boundary)	Unit	P2	3	Pass
TC-U17	FR1	English verbs get the right Bangla ending for 15 tense/person forms	Unit	P1	15	Pass
TC-U18	FR1	Base verb of English forms, including -ie verbs (dies, died)	Unit	P1	16	Pass
TC-U19	FR1	Bangla place ending -এ / -তে / -য় / -য়ে (বইয়ে)	Unit	P1	8	Pass
TC-U20	FR1	Bangla possessive ending -র / -ের / -য়ের (ভাইয়ের)	Unit	P1	8	Pass
TC-U21	FR1	Imperatives, verbal nouns and fixed English phrases	Unit	P2	2	Pass
TC-U22	FR1	Noakhali "will not" rule: খাবো না = খাইতান ন, both ways	Unit	P1	8	Pass
TC-D01	NFR4	Every word list row is complete, without stray spaces, with a known status (25 lists)	Data	P1	26	Pass
TC-D02	NFR4	No certain typing mistakes (same vowel sign twice, split ো)	Data	P1	26	Pass
TC-D03	NFR4	No duplicate rows in any word list	Data	P2	26	Pass
TC-D04	NFR4	Words approved by a native speaker are written in valid Bangla script	Data	P1	27	Pass
TC-D05	NFR4	The verified list curated.csv is clean	Data	P1	1	Pass
TC-01	FR1, FR3	Native-speaker sentences translate exactly (Standard ↔ Noakhali, Standard → Chattogram)	Integration	P1	17	Pass
TC-02	FR1, NFR4	Approved Noakhali, Chattogram and Garo words are used both ways	Integration	P1	17	Pass
TC-03	NFR4	The three native-speaker-reviewed lists are loaded with their alternatives	Integration	P1	4	Pass
TC-04	FR1	English sentences follow Bangla grammar (I will eat rice → আমি ভাত খাবো)	Integration	P1	19	Pass
TC-05	FR3	A text of several sentences is matched sentence by sentence	Integration	P1	1	Pass
TC-06	FR4	Every word gets a word box	Integration	P1	1	Pass
TC-07	FR1	An unknown language is rejected	Integration	P1	1	Pass
TC-08	FR5	A word added on the Words screen is used by the translator, then deleted	Integration	P1	2	Pass
TC-09	FR3, FR5	A saved phrase is found as an exact sentence	Integration	P2	1	Pass
TC-10	NFR3	Rate limit: 2 and 3 of 3 requests allowed, the 4th blocked; the XAMPP computer is never limited	Integration	P1	4	Pass

TC ID	Req ID	Title / Test Case	Level	Priority	Runs	Result
TC-11	FR8	Register → me → log out → me → log in again (email in capitals)	Integration	P1	1	Pass
TC-12	FR8, NFR3	Registration input classes and boundaries (password 7/8, email 150/151, empty name, list instead of text)	Integration	P1	11	Pass
TC-13	FR8	A second account with the same email is refused (409)	Integration	P1	1	Pass
TC-14	FR8	Login with empty fields gets 400	Integration	P2	1	Pass
TC-15	FR1	Translate API gives the native-speaker sentence	Integration	P1	1	Pass
TC-16	FR2	Invalid translate input gets 400 with a message (no text, unknown language, list)	Integration	P1	8	Pass
TC-17	FR2	Text length 1, 499, 500, 501 and 3,000 characters: at most 500 translated, never refused	Integration	P1	5	Pass
TC-18	FR8	A logged-in user's translation is saved in History and can be deleted	Integration	P1	1	Pass
TC-19	FR5	Words screen: add, search, edit, translate with, and delete a word	Integration	P1	1	Pass
TC-20	FR5	Word and meaning length 100 / 101 letters, empty values, unsupported language	Integration	P1	7	Pass
TC-21	FR5, NFR3	Words page number kept between 1 and 100,000; % and _ are searched as letters	Integration	P2	8	Pass
TC-22	NFR3	User B cannot delete user A's history; a guest cannot read it	Integration	P1	2	Pass
TC-23	FR2	ো typed as ে + া gives the same translation and is found in Words	Integration	P1	1	Pass
TC-24	FR2, FR1	Banglish full stop becomes ।; "Mr." does not end an English sentence	Integration	P2	2	Pass
TC-25	FR7	A lesson answer needs a logged-in user (401)	Integration	P1	1	Pass
TC-26	FR8	Profile: spoken language, district and dialect are saved; unknown values refused	Integration	P2	8	Pass
TC-27	FR5	Word contributions: valid, missing meaning, unsupported dialect, logged-in user	Integration	P2	5	Pass
TC-28	NFR3	Wrong HTTP methods get 405; invalid history ids get 400	Integration	P2	6	Pass
TC-29	FR8	Word of the day works without a language (Chattogram)	Integration	P2	1	Pass
TC-35	FR1, FR4	Dialect to dialect (Noakhali ↔ Chattogram): word boxes and the Bridge show the target dialect word; dialect → Standard unchanged	Integration	P1	3	Pass
TC-36	FR7, NFR1	Learning instructions in the learner's language: Standard Bangla for Bangla varieties, reviewed words or symbols for Garo and Mro, never English; unknown language refused	Integration	P1	5	Pass
TC-37	FR7, NFR4	Word questions show the learner's own word only from a native-speaker reviewed list (Garo ছি = পানি; none for Mro)	Integration	P1	1	Pass
TC-30	FR1, FR3	Translation system cases: every direction, exact sentence first, English grammar (33 cases)	System	P1	33	Pass
TC-31	FR2, FR3	Input validation and limits (8 cases)	System	P1	8	Pass
TC-32	FR5, FR6	Dictionary browse, search, edit log by address and administrator undo (19 cases)	System	P1	19	Pass
TC-33	FR8	Accounts, history, quiz and word of the day (27 cases)	System	P1	27	Pass
TC-34	FR7	Learning path and graded lessons (10 cases)	System	P1	10	Pass
E2E-01	FR1, FR2, FR4	Phone browser: translate, voice, Listen, Scan and word boxes (UI-01–UI-15, UI-40)	E2E	P1	16	Pass
E2E-02	FR5	Phone browser: Words screen add, edit, delete, XSS shown as text (UI-16–UI-22)	E2E	P1	7	Pass
E2E-03	FR7, FR8	Phone browser: history, profile, Learn path and quiz (UI-23–UI-30, UI-38)	E2E	P1	9	Pass
E2E-04	NFR1	Layout at 360–414 px, install, offline start, long words (UI-31–UI-37, UI-39)	E2E	P1	8	Pass
E2E-05	FR4, FR7	Phone browser: Bridge shows the word pairs, dialect-to-dialect boxes, Mro learner sees no English (UI-41–UI-43)	E2E	P1	3	Pass
PT-01	NFR2	One user and 20 users: translation fast, no errors (LT-01, LT-02)	Performance	-	2	Pass
PT-02	NFR2	50–300 users and a 500-request spike: slower but no errors (LT-03–LT-06)	Performance	-	4	Fail
PT-03	NFR2	3-minute soak, memory, recovery, MySQL connections (LT-07–LT-10)	Performance	-	4	Pass
PT-04	NFR2	Page speed on fast and slow 4G, repeat visit, click to result (PF-05–PF-08)	Performance	-	4	Pass
SEC-01	NFR3	Password guessing: blocked after 8, 2,000-attempt flood, faked addresses (BF-01–BF-07)	Security	P1	7	Pass
SEC-02	NFR3	Flooding edits, sign-ups, contributions and voice (FL-01–FL-04)	Security	P1	4	Pass

TC ID	Req ID	Title / Test Case	Level	Priority	Runs	Result
SEC-03	NFR3	SQL injection (117 strings × 13 fields), XSS, JSON type, mass assignment (IN-01–IN-07)	Security	P1	7	Pass
SEC-04	NFR3	Sessions and passwords: logout, fixation, bcrypt, weak passwords, enumeration (SS-01–SS-05)	Security	P1	5	Pass
SEC-05	NFR3	Access control: admin pages, path traversal, hidden files, TRACE (AC-01–AC-06)	Security	P1	6	Pass
SEC-06	NFR3	Security headers, error leaks, fuzzing, CSRF, CORS, clickjacking (HD, FZ, BR)	Security	P1	10	Pass
SEC-07	NFR3	Static code checks: syntax, dangerous functions, SQL, escaping, secrets, locks (CS-01–CS-09)	Security	P2	9	Pass
REL-01	FR5, FR6	40 people editing at once; edits survive a reset; deletes (RL-01–RL-05)	Reliability	P1	5	Pass
REL-02	NFR2	MySQL stops and starts; Apache restarts (RL-06–RL-09)	Reliability	P1	4	Pass
A11Y-01	NFR1	WCAG 2.1 AA on 10 screens, keyboard, 200 % zoom (AX-01–AX-04)	Accessibility	-	4	Pass
A11Y-02	NFR1	6 screen sizes, dark mode, reduced motion (DV-01–DV-03)	Accessibility	-	3	Pass

Summary. Total test cases: 84. Pass: 83. Fail: 1. Behind them are 673 individual runs.

5.2 Detailed test cases

Six of the most important cases:

ID	Requirement	Setup	Steps	Expected	Result
TC-01	FR1, FR3: native sentence translates exactly	Database loaded; Noakhali native sentence pair	Translate আমি ভাত খাবো না to Noakhali, and back	আইঁ ভাত খাইতান ন / আমি ভাত খাবো না	Pass
TC-U01	FR8, NFR3: password length	none (unit)	vs_password_problem() with 0, 7, 8, 9, 199, 200, 201 characters and 7/8 Bangla letters	Refused below 8 and above 200 letters	Pass
TC-12	FR8: registration boundaries	App on Apache	Register with a valid 150- and 151-character email	200, then 400 "Email address is too long"	Pass
TC-22	NFR3: users cannot touch each other's data	Users A and B; A has a history entry	B deletes A's entry by its id; guest reads history	Not deleted; B and the guest see nothing	Pass
TC-D04	NFR4: approved words are valid Bangla	All word lists	Check every approved word for a vowel sign on a vowel letter	None (8 approved as written by the reviewer)	Pass
SEC-01	NFR3: password guessing	App on Apache	2,000 logins from 40 attackers; faked addresses	Blocked after 8; no errors; others keep translating	Pass

5.3 Boundary value analysis

Each input is tested at and just beside its limits. Test numbers refer to the list in 5.1.

Input	Valid	Boundaries tested	Tests
Password	8 to 200 letters	0, 7, 8, 9, 199, 200, 201; 7 and 8 Bangla letters; 200 Bangla letters	TC-U01, TC-12
Email	valid address, at most 150 characters	150, 151 (part before @ at most 64)	TC-12
Translate text	1 to 500 characters (more is cut)	0, only spaces, 1, 499, 500, 501, 3,000	TC-16, TC-17
Word / meaning	1 to 100 letters	0, 100, 101	TC-20
Words page	1 to 100,000	0, -5, text, 1, 100,000, 100,001, 10^15	TC-21
Rate limit	max requests per window	max-1, max, max+1	TC-10
Verb variant	words longer than 2 letters	2, 3, 4 letters	TC-U08
Fuzzy key	any length	0, 1, 2 tokens	TC-U14, TC-U16
Visitor address header	up to 64 characters	64, 100	TC-U04

5.4 Word review states

needs_native_review → native_speaker_reported (main form) or native_speaker_alternative (other spelling of the same Standard word). Only reported and alternative words are shown as verified; only main forms are used when translating into a dialect. We test each state (TC-02, TC-03, TC-D04) and the wrong ones: an unreviewed word used as trusted (TC-D04), an odd unreviewed spelling marked as approved (TC-D04), an edited word keeping its old meaning (TC-08).

6. Test Execution Results

6.1 Summary

Level	Runs	Passed	Failed	Pass %
Unit	202	202	0	100.0%
Data	106	106	0	100.0%
Integration	147	147	0	100.0%
System	97	97	0	100.0%
E2E	43	43	0	100.0%
Performance	14	10	4	71.4%
Security	48	48	0	100.0%
Reliability	9	9	0	100.0%
Accessibility	7	7	0	100.0%
Total	**673**	**669**	**4**	**99.4%**

PHPUnit in numbers: 421 tests and 43,985 assertions (Unit 202, Data 106, Integration 33, HttpFlows 80), 0 failures. The data suite has many assertions because it checks every row of 25 word lists. The GitHub CI pipeline runs PHPUnit with coverage, the code checks and the functional suites on every push.

6.2 Test run evidence

Each suite was run in a terminal from the project folder on the tested build. Per-test names come from PHPUnit --testdox and the PASS/FAIL lines of our Python suites.

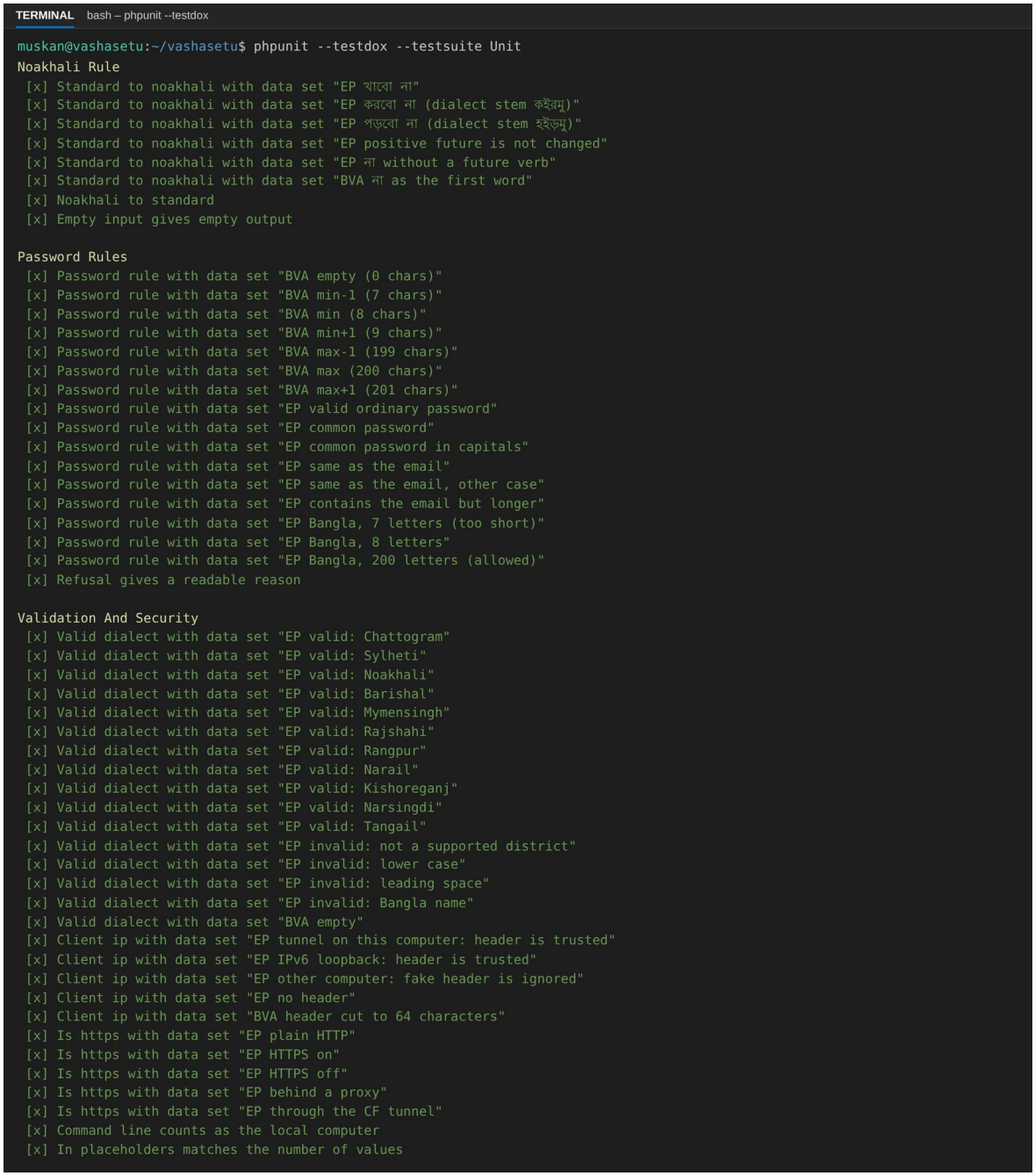


Figure 1: PHPUnit, all four suites (TestDox, part of the list)

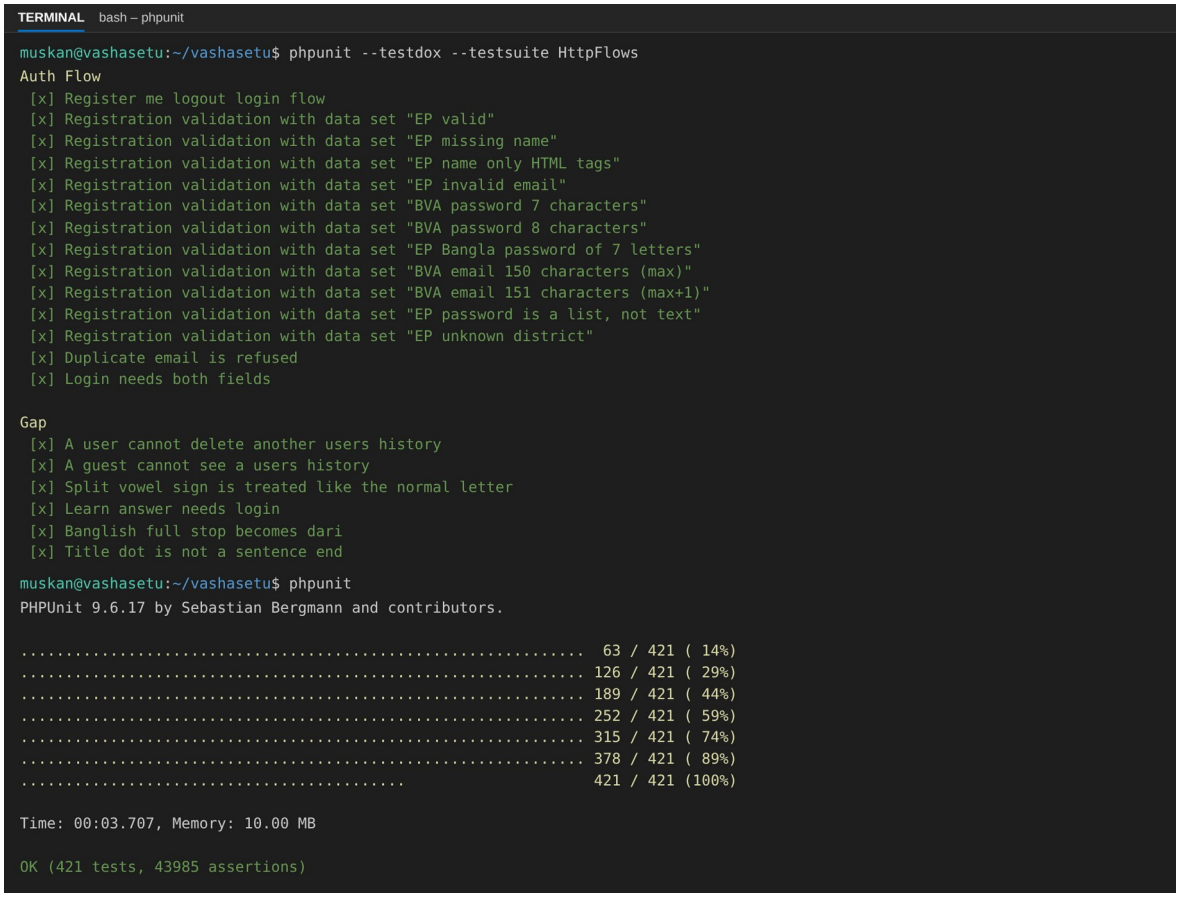


Figure 2: PHPUnit summary and the data suite

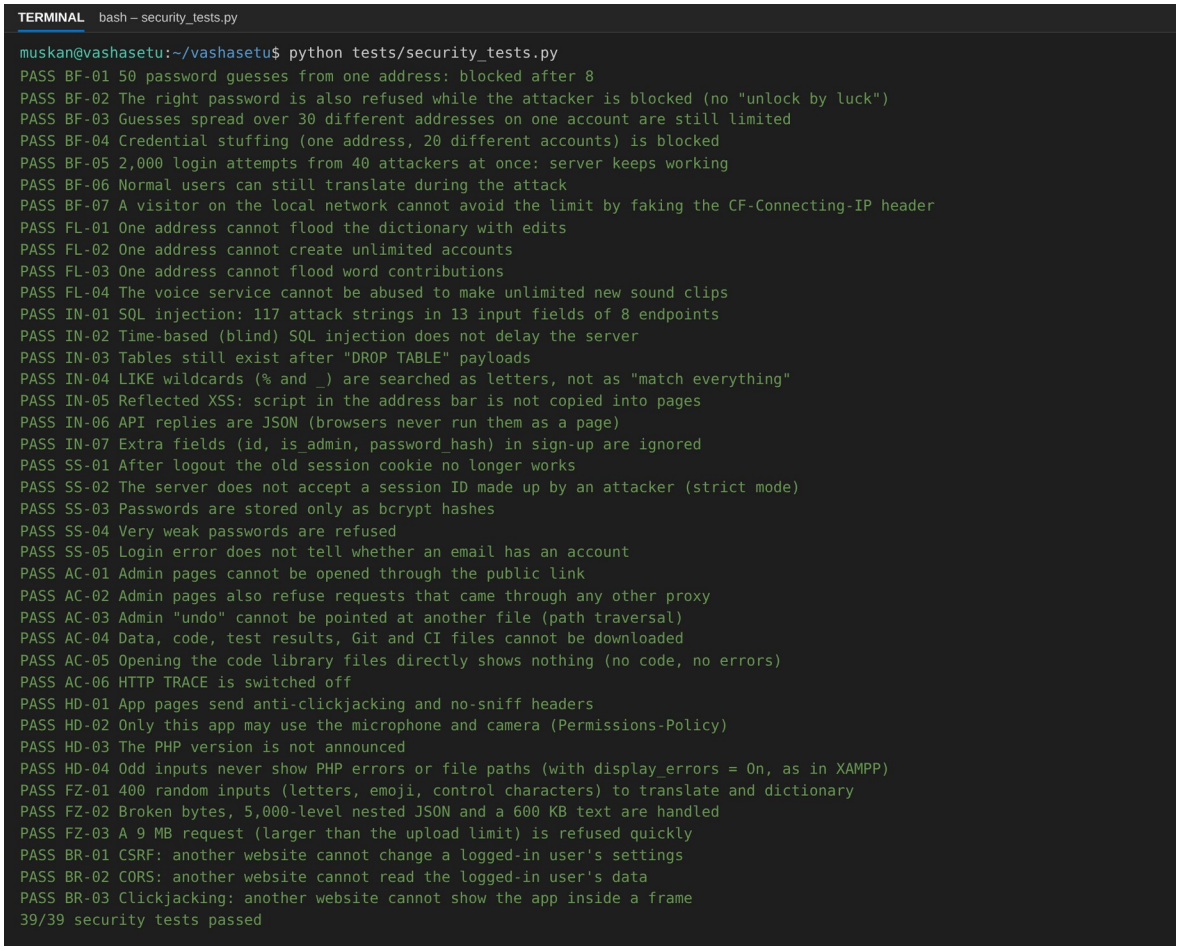


Figure 3: Security tests (39 passed)

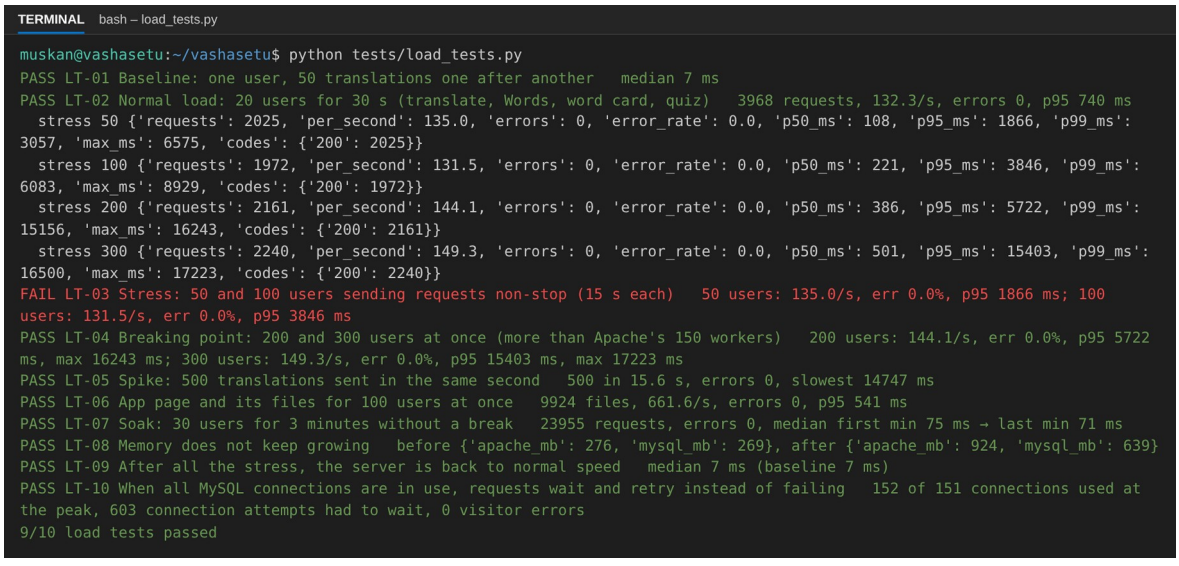


Figure 4: Load, stress, spike and soak tests (10 passed)

```
TERMINAL  bash – system / ui / reliability / a11y

muskan@vashasetu:~/vashasetu$ python tests/system_tests.py
PASS SE-12 config.php does not show the database password
PASS PF-01 40 translations: average time under 1 second
PASS PF-02 Dictionary page loads in under 1 second
PASS PF-03 30 translations from 10 users at the same time
PASS PF-04 Home page HTML loads quickly
115/115 system tests passed

muskan@vashasetu:~/vashasetu$ python tests/ui_tests.py
PASS UI-35 App screen still opens with no internet (offline)
PASS UI-37 A very long word (100 letters, no spaces) wraps inside its box
PASS UI-38 When a quiz has too little data, a message is shown on the page (no pop-up box)
PASS UI-43 Instructions follow the learner's language: a Mro speaker sees pictures and symbols, no English
PASS UI-36 No JavaScript errors during the whole test run
43/43 UI tests passed

muskan@vashasetu:~/vashasetu$ python tests/reliability_tests.py
PASS RL-01 40 people add a word at the same moment: no edit is lost from the trusted list
PASS RL-02 The edit log stays readable when 40 edits arrive together (no mixed lines)
PASS RL-03 Database and files agree after the parallel edits
PASS RL-04 Dictionary edits survive setup.php?reset=1 (they are re-applied from user_edits.csv)
PASS RL-05 Deleting the 40 test words removes them everywhere
PASS RL-06 MySQL stopped: the API answers at once with a clear message
PASS RL-07 MySQL stopped: the app page itself still opens
PASS RL-08 After MySQL starts again the app works at once, without restarting Apache
PASS RL-09 A logged-in user stays logged in after the web server restarts
9/9 reliability tests passed

muskan@vashasetu:~/vashasetu$ python tests/a11y_perf_tests.py
PASS AX-01 WCAG 2.1 AA check (axe-core) on 10 screens: no critical or serious problems   none
PASS AX-02 WCAG 2.1 AA check: moderate and minor problems   none
PASS AX-03 The translator can be used with the keyboard only (Tab reaches the main controls, focus is visible)   19 different
controls; focus visible on 22/25; text box reached: True
PASS AX-04 Text zoomed to 200 % still fits (no sideways scrolling)   0 px
PASS PF-05 First visit on a fast connection: page shows quickly   FCP 196 ms, LCP 0 ms, CLS 0.000, 13 files, 253 KB
PASS PF-06 First visit on a slow phone network (1.6 Mbit/s, 150 ms, slow CPU ×4)   FCP 1496 ms, LCP 1496 ms, fully loaded 1590
ms, 253 KB
PASS PF-07 Second visit on the slow network is much faster (files kept by the offline helper)   8 KB downloaded (first visit
253 KB), FCP 340 ms
PASS PF-08 Time from pressing Translate to seeing the result   451 ms, 947 ms, 967 ms
PASS DV-01 Works on 6 screen sizes from a 320 px phone to a 1366 px laptop, and in landscape   all fine
PASS DV-02 Phone set to dark mode: text stays readable   contrast 14.1 : 1
PASS DV-03 Phone set to "reduce motion": endless animations stop   0 endless animations
11/11 accessibility/speed/device tests passed
```

Figure 5: System, UI, reliability and accessibility suites

```
TERMINAL  bash – mutation_check.py

muskan@vashasetu:~/vashasetu$ python tests/mutation_check.py
KILLED   M-01 Password minimum 8 → 7
KILLED   M-02 Password maximum 200 → 199
KILLED   M-03 Email check becomes case-sensitive
KILLED   M-04 Length counted in bytes again (defect D28)
KILLED   M-05 Tunnel header trusted from anyone (defect D11)
KILLED   M-06 Tangail dropped from the dialect list
KILLED   M-07 Verb-variant boundary 2 → 3 letters
KILLED   M-08 Split ◌া no longer joined (defect D31)
KILLED   M-09 Cost of the optional inherent vowel
KILLED   M-10 Aspirate similarity K/KH
KILLED   M-11 Place ending after ি/ু
KILLED   M-12 Possessive after a vowel letter (defect D25)
KILLED   M-13 Future ending for I
KILLED   M-14 -ie verbs (died) not found (defect D27)
KILLED   M-15 Titles end sentences again (defect D30)
KILLED   M-16 Banglish full stop not converted (defect D32)
KILLED   M-17 Trailing space/line break of a sentence lost
KILLED   M-19 Rate limit lets one extra request through
KILLED   M-20 Empty key allowed in fuzzy look-ups
KILLED   M-21 Place ending after ◌া (ঢাকায়)
KILLED   M-22 Similarity of য/জ sounds
KILLED   M-23 Long texts no longer translated sentence by sentence
SURVIVED M-24 Irregular form of an unknown verb returned (defect D26)
KILLED   M-18 Every password refused
Mutation score: 23/24 mutants killed
```

Figure 6: Mutation testing: planted bugs and the result

6.3 Failed tests

Test	Why it fails	Defect
PT-02	LT-03: with 100 users sending requests non-stop, 95% of replies took 3.85 s, over our own 3 s target (0 errors). The test computer was restarted during the session and now answers about 132 requests per second instead of 190–215 in earlier runs of the same code; LT-04 to LT-06 (200–300 users, spike) still had no errors.	D34

The test stays in the suite so the gap is visible. Every other test case passes.

7. Code Coverage

Layer	PHPUnit only	All suites	Target
Data access, config and security	30.6% (55/180)	85.6% (154/180)	
Translation engine and helpers	76.9% (1247/1621)	91.1% (1476/1621)	60%
API endpoints	0.0% (0/754)	87.8% (649/739)	
Pages and setup	0.0% (0/259)	93.7% (238/254)	
Whole PHP code	**46.3%**	**90.1% (2517/2794)**	60%

Lowest files: api/learn.php 77.0%, api/tts.php 78.8%, api/edit_lib.php 81.8%. The lines not covered are catch blocks for a database failure in the middle of a request, the Google voice download (no internet on the test computer), settings that are switched off (login-only editing), and one dead branch in api/word.php found by the coverage report. Coverage of the PHP code is recorded both inside PHPUnit and for every web request during the HTTP and Python suites (tests/phpunit/coverage/prepend.php), and merged into one report (tests/results/coverage/html/index.html). The JavaScript front end is covered by the E2E tests only.

Mutation testing. To check that the tests assert the right behaviour and do not only run the code, tests/mutation_check.py planted 24 small bugs one at a time (for example "< 8" became "< 7", or a fixed defect was put back). 23 of 24 were caught. At first 4 survived; we added tests for 3 of them. The last one cannot be seen from outside with the current data (an equivalent mutant).

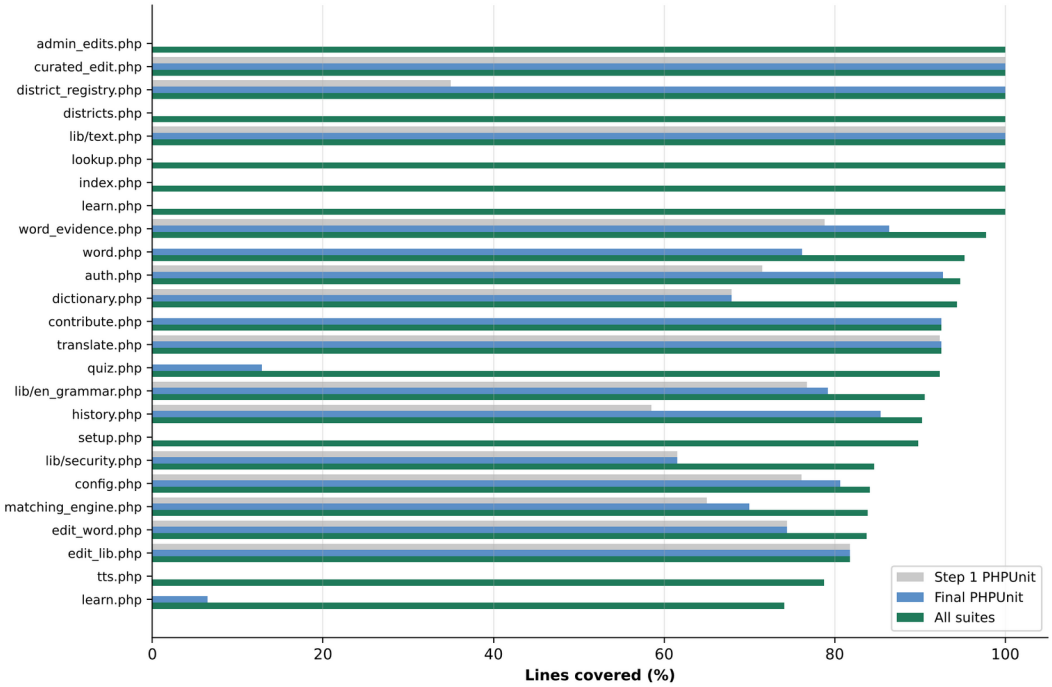


Figure 7: Line coverage by file – Step 1 PHPUnit suite, final PHPUnit suite, all suites

8. Non-Functional Testing

8.1 Performance

Load	p50	p95	Errors	Result
20 users, mixed requests	47 ms	740 ms	0/3,968	Pass
50 users, mixed requests	108 ms	1,866 ms	0/2,025	Pass
100 users, mixed requests	221 ms	3,846 ms	0/1,972	Pass
200 users, mixed requests	386 ms	5,722 ms	0/2,161	Pass
300 users, mixed requests	501 ms	15,403 ms	0/2,240	Pass
500 translations at once (spike)	–	last after 15.6 s	0/500	Pass
30 users for 3 minutes (soak)	–	drift ×0.95	0/23,955	Pass

Users sent requests without any pause (60% translations, 20% Words pages, 10% word cards, 10% quizzes) on 2 CPU cores. Above 150 users – Apache's worker limit – requests wait in a queue and get slower, but none fail. The SRS asks for a translation in under 1 second for normal use: a single translation takes a median of 7 ms. Page speed: first content after 196 ms on a fast connection and 1496 ms on slow 4G (Google's "good" limit is 1,800 ms).

8.2 Security

Check	How	Result
Password guessing	Blocked after 8 tries per address/account; 2,000-attempt flood; faked addresses (SEC-01)	Pass
Flooding	Limits on edits, sign-ups, contributions, voice (SEC-02)	Pass
Access control	User B tries A's history (TC-22); admin pages through the public link; path traversal; hidden files (SEC-05)	Pass
SQL injection	' OR 1=1 -- and 116 more strings in 13 fields; time-based; LIKE wildcards (SEC-03, TC-21)	Pass
XSS	Script in inputs and the address bar shown as text (SEC-03, E2E-02)	Pass
Sessions	Logout ends the session; made-up session id refused; bcrypt hashes (SEC-04)	Pass
CSRF, CORS, clickjacking	Evil page in a real browser (SEC-06)	Pass
Error leaks	No PHP warnings or paths even with display_errors On (HD-04)	Pass
Secrets in code	Static scan of PHP and JS (CS-08)	Pass

8.3 Static analysis

php -l on all PHP files, node --check on the JavaScript and py_compile on the Python scripts: 0 errors. code_scan.py also checks that no dangerous PHP functions are used, that no SQL is built from user input, that page output is escaped and that shared files are written under a lock (9 of 9 checks pass).

8.4 Accessibility and devices

Screen	axe-core WCAG 2.1 AA problems
splash	0
home	0
translate	0
dictionary	0
history	0
profile	0
quiz	0
login	0
register	0
learn	0

Keyboard: the translator can be used with Tab only; text zoomed to 200% needs no sideways scrolling; works on 6 screen sizes from 320 to 1366 px, in dark mode (contrast 14:1) and with "reduce motion" (A11Y-01, A11Y-02).

8.5 Usability

Not done as a formal task test yet. Native speakers who reviewed the word lists used the app to check translations. Plan: three students from other sections translate a sentence, add a word and finish one lesson; we will record completion and time.

9. Defect Report

9.1 Summary by severity

S1 – data loss or a working attack on accounts; S2 – wrong result or a feature fails; S3 – fails only for some inputs or a hardening gap; S4 – code or test quality.

Severity	Found	Fixed	Open
S1	2	2	0
S2	21	21	0
S3	11	10	1
S4	4	4	0
Total	**38**	**37**	**1**

9.2 Defect log

ID	Problem	Sev	Found by	Status	Regression test
D1	An unknown input or output language returned "500 Translation failed" instead of a clear 400 message.	S2	TV-03, TV-04	Closed	TV-03, TV-04
D2	Sending a list where text is expected (e.g. password ["…"]) stopped PHP with an empty 500 reply, on every API.	S2	TV-08, ED-08, AU-07	Closed	TV-08, ED-08, AU-07
D3	Noakhali → another dialect showed the Standard Bangla bridge (আমি ভাত খাবো) instead of the other dialect (আঁই ভাত খাইয়ুম). The old test only checked that some text came back.	S2	TR-08	Closed	TR-08
D4	A very long word (no spaces) ran out of its dictionary row (1,779 px) and out of the result card (992 px) on a 360 px phone.	S2	Exploratory → UI-37	Closed	Exploratory → UI-37
D5	Learn showed problems in a browser pop-up (alert), which blocks the phone screen.	S3	Exploratory → UI-38	Closed	Exploratory → UI-38
D6	After opening the app with no internet, a logged-in person stayed a "guest" (Learn asked to log in) until the page was reloaded.	S3	Found while running UI-38 → UI-39	Closed	Found while running UI-38 → UI-39
D7	Two button actions (stopMic, dictMore) were defined twice in app.js.	S4	Code review	Closed	Code review
D8	Regression test T17 used a fixed account; a second run within 15 minutes failed because the login limit blocked it.	S4	Regression run	Closed	Regression run
D9	English sentences that are not in the phrase list were translated word by word in English order: "I will eat rice" → আমি অভিলাষ খাওয়া ভাত instead of আমি ভাত খাবো. English → a dialect gave no result at all.	S2	TR-19	Closed	TR-19
D10	No limit on dictionary edits, contributions or new accounts: one script could add thousands of words or accounts in minutes.	S2	FL-01–FL-03	Closed	FL-01–FL-03
D11	The server trusted the CF-Connecting-IP header from anyone, so a visitor could name a new address on every guess and never be blocked.	S1	BF-07	Closed	BF-07
D12	CSRF: a form on another website could send changes to the API as the logged-in user; in the test it even logged the user out.	S2	BR-01	Closed	BR-01
D13	No anti-clickjacking headers: another website could show the app in a hidden frame.	S2	BR-03, HD-01	Closed	BR-03, HD-01
D14	Very weak passwords (6 characters, "123456", the email itself) were accepted.	S2	SS-04	Closed	SS-04
D15	The server accepted a session id made up by an attacker (session fixation risk).	S3	SS-02	Closed	SS-02
D16	tests/ (with results) and .github/ could be downloaded.	S4	AC-04	Closed	AC-04
D17	PHP announced its version, and with XAMPP's display_errors = On a warning could show file paths.	S4	HD-03, HD-04	Closed	HD-03, HD-04
D18	A very large Words page number (page=10^15) gave an empty 500 error.	S3	Fuzzing	Closed	Fuzzing
D19	Persistent MySQL connections: each Apache worker kept a connection for ever, so under load all 151 connections were used and new requests failed.	S2	LT-10	Closed	LT-10

ID	Problem	Sev	Found by	Status	Regression test
D20	With 200–300 users, 14–21% of requests got 503 because finished connections were still closing when new ones started.	S2	LT-04 (second run)	Closed	LT-04 (second run)
D21	40 people adding words at the same moment could wipe the trusted word list (curated.csv: 81 rows → 5).	S1	RL-01	Closed	RL-01
D22	Noakhali rule: পড়বো না stayed পড়বো না instead of হইড়তান ন.	S2	UT-11	Closed	UT-11
D23	Accessibility: the clickable map was marked as a picture, the dialect list in Profile had no label, and some Learn text had too little contrast.	S3	AX-01, AX-02	Closed	AX-01, AX-02
D24	After the whole Noakhali word list was approved, আমি had three approved spellings (আইঁ, আঁই, আই) and word-by-word translation picked আই, so আমি ভাত খাবো became আই ভাত খামু. The Words screen also hid approved alternative spellings.	S3	TR-17 (after the native-speaker review)	Closed	TR-17 (after the native-speaker review)
D25	English place/possessive endings after a vowel letter gave invalid Bangla (বইে, ভাইের)	S2	TC-U19, TC-U20	Closed	TC-U19, TC-U20
D26	"I am lying" stopped PHP with a TypeError; the user saw a wrong error message	S2	TC-U18	Closed	TC-U18
D27	-ie verbs (dies, died) not recognised	S3	TC-U18	Closed	TC-U18
D28	Password length counted in bytes: 7 Bangla letters accepted, 200 refused	S2	TC-U01	Closed	TC-U01
D29	Email longer than 150 characters gave 500 instead of 400	S3	TC-12	Closed	TC-12
D30	"Mr."/"Dr." ended a sentence: মি। Karim এখানে আছে।	S2	TC-U10, TC-24	Closed	TC-U10, TC-24
D31	ো typed as ে + া not recognised: wrong translation, nothing found in Words	S2	TC-U07, TC-23	Closed	TC-U07, TC-23
D32	Banglish "." stayed "." instead of " ।"	S3	TC-U12, TC-24	Closed	TC-U12, TC-24
D33	34 typing mistakes, 1 duplicate row, 2 invalid approved spellings in the word lists	S2	TC-D02–TC-D04	Closed	TC-D02–TC-D04
D34	With 100 users non-stop, 95% of replies take more than 3 s (3.85 s in this run) on the 2-core test computer (no errors)	S3	PT-02 (LT-03)	Open	PT-02 (LT-03)
D35	The Bridge screen stayed empty after a translation (JavaScript error "pairShown is not defined")	S2	E2E-05 (UI-41)	Closed	E2E-05 (UI-41)
D36	Noakhali → Chattogram: word boxes and the Bridge showed the Standard word in between (খাবো) instead of the Chattogram word (খাইয়ুম)	S2	TC-35, UI-42	Closed	TC-35, UI-42
D37	Learning page instructions were in English for every learner, also for Mro and Garo speakers	S2	TC-36, UI-43	Closed	TC-36, UI-43
D38	Word questions showed the answer's spelling in English letters (ma, pani) before answering	S3	Found in use; checked by hand	Closed	Found in use; checked by hand

D1–D24 were found by the system, UI, security, load, reliability and accessibility suites; D25–D33 by the PHPUnit suite, the generated-test review and the gap analysis; D34 by the load test; D35–D38 by a user while using the app on a phone; D35–D37 are now covered by new tests (TC-35–TC-37, UI-41–UI-43) and D38 was checked by hand. Every fixed defect has a test that failed before the fix and passes after it.

9.3 Main defects

ID	Test	Observed	Cause	Fix
D21	REL-01	40 parallel edits: curated.csv went from 81 rows to 5	Each edit read, changed and rewrote the file without a lock	File lock, write to a temp file, replace in one step
D11	SEC-01	Unlimited password guesses with a new fake address each time	CF-Connecting-IP trusted from anyone	Header trusted only from the tunnel on this computer
D19	PT-03	151/151 MySQL connections in use; requests failed	Persistent PDO connections held by every Apache worker	Normal connections; wait and retry when MySQL is full (D20)
D26	TC-U18	"I am lying" → 400 "Invalid input…"	"lying" mapped to the verb lie, which has no Bangla forms → TypeError	Such forms are ignored; removed from the list
D30	TC-U10 (generated G-01)	মি। Karim এখানে আছে।	The full stop of "Mr." was taken as a sentence end	Titles and list numbers stay with the next words
D31	TC-23	আমি ভাত খাবো typed with ে + া → আঁই ভাত খাবো	Text compared byte by byte, not as Unicode letters	bn_compose() joins split vowel signs (same as NFC)

10. Tool-Assisted Testing Log

The PHPUnit tests and scripts were written with the help of an AI assistant (Claude) under our direction, and it also generated the 25 edge-case tests of Step 2 (Qodo needs its IDE plugin and an account, and Diffblue Cover only supports Java). Every expected value was checked by us against the specification, native-speaker data or Bangla grammar. What the tool got wrong:

Test	What the tool got wrong	What we did
TC-U10 (G-01, G-02)	Expected "Dr. Rahim came." to be two sentences – it copied the buggy output	Corrected the expectation; fixed the bug (D30)
TC-U10 (G-03)	Expected a list number "১." to be a sentence of its own	Expectation changed to the rule after the fix
TC-U14	Expected distance 1.5 for vowel vs consonant; delete + insert (1.3) is cheaper	Fixed the expected value; added a substitution-cost test
TC-12	Built a 150-character email with a 130-character part before @ (invalid; max 64)	Made the domain longer; the valid address then found D29
TC-25	Sent the learning action in the URL; the API reads it from the JSON body	Fixed the request
TC-29	Expected 400 without a language; the code uses Chattogram (the 400 branch is dead code)	Test checks the real behaviour; dead code noted in section 7
Mutation run	4 planted bugs not caught (weak assertions)	Added 3 tests; 1 is an equivalent mutant

Summary of the generated tests: 25 generated, 22 correct, 2 wrong, 1 questionable (tests/phpunit/generated_review.json).

11. Test Completion Summary

11.1 Exit criteria evaluation

All 5 exit criteria of section 3.2 are met. Not met or not done yet: the 3 s target at 100 non-stop users (D34, by 0.85 s), Safari/Firefox, and a formal usability test.

11.2 Residual risks and known issues

Risk	Impact on users	What to do
Most dialect words not reviewed yet	Unreviewed words may be wrong; they are labelled "needs native review"	Continue the native reviews for the other dialects
8 approved Chattogram words with unusual script	They show a dotted circle on some phones	Ask the reviewer for the intended spelling
Open dictionary editing	Anyone with the link can edit (logged, limited, undoable)	Set REQUIRE_LOGIN = true for public use
Only Chromium tested	Unknown layout issues on Safari/Firefox	Check by hand before wider release
Slower replies above about 100 users at once (D34)	Replies can take 3 s or more, but do not fail	For a bigger audience: more CPU cores and Apache workers
Login block can hit the real owner	Up to 15 minutes blocked during an attack	Accepted trade-off

11.3 Release Recommendation

Go with known issues. 83 of 84 test cases pass (the one failure is a 0.85 s miss of our own load target, D34), both S1 defects (word list wiped by parallel edits, login limit bypass) are fixed, and the reviewed word lists are used correctly. Before wider public use: require login for editing and finish the native reviews of the most common words.

11.4 Lessons learned

- Testing at the boundaries with Bangla input (letters, not bytes; vowel letters; split vowel signs) found bugs that English test data never shows.
- A test generator that copies the current output locks bugs in; every generated expected value has to be checked.
- The data is part of the product: a data test found 37 errors in the word lists that no code test could see.
- Coverage shows what is run, mutation testing shows what is really checked; both together gave us confidence.

12. Individual Contribution

Member	Test cases	Defects found	Defects fixed
Shobnom Sultana Muskan	All (84)	38	37
Sadia Akter Liza	Manual testing on phones (all screens)	–	–

13. Sign-off and Declaration

I declare that this report describes testing done on the stated version. Results can be reproduced from the project files with the commands in section 2.2. Tool use is disclosed in section 10.

	Name	Signature	Date
Prepared by	Shobnom Sultana Muskan		31 Oct 2026
Prepared by	Sadia Akter Liza		31 Oct 2026

Appendix A: Evidence Index

All in tests/results/ of the project.

#	Evidence	File
A1	PHPUnit results	phpunit-junit.xml, phpunit-testdox.txt
A2	System, UI, security, load, reliability, accessibility results	system_results.json, ui_results.json, security_results.json, load_results.json, reliability_results.json, a11y_perf_results.json, screens/
A3	Coverage	coverage/html/index.html, coverage/clover.xml, coverage/summary.txt
A4	Mutation testing	mutation_results.json
A5	Generated-test review	tests/phpunit/generated_review.json
A6	Code checks and unit checks	code_scan_results.json, unit_results.json
A7	Native-speaker reviewed lists	Google Sheets Vashasetu_Noakhali_Words, Vashasetu_Garo_Words, Vashasetu_Chattogram; database/languages/*/words.csv

Appendix B: Pre-Submission Checklist

Yes: version stated and all results come from it

Yes: tests run from the commands in section 2.2

Yes: CI runs PHPUnit with coverage on every push

Yes: every FR and NFR has a test

Yes: coverage numbers match the report files

Yes: no open S1 defects; every fixed defect has a regression test

Yes: tool use disclosed

Yes: no real personal data or secrets in tests (…@example.com accounts only)